F18CF – Linear Algebra

Tutorial Sheet 2: Vectors of $\mathbb{R}^n$

Getting started.

1. (a) Write the vector $(5,3)^T$ of $\mathbb{R}^2$ as a linear combination of the vectors $(1,-1)^T$, $(1,1)^T$.

   (b) Write the vector $(5,3,1)^T$ of $\mathbb{R}^3$ as a linear combination of the vectors $(1,-1,0)^T$, $(1,0,-1)^T$, $(0,1,1)^T$.

2. (a) Does the vector $(5,3)^T$ lie in the span of the vectors $(1,2)^T$ and $(1,3)^T$?

   (b) Does the vector $(5,3,1)^T$ lie in the span of the vectors $(1,2,1)^T$, $(1,3,1)^T$ and $(1,4,1)^T$?

   (c) For which $a \in \mathbb{R}$ does the vector $(-2,1,a)^T$ lie in the span of $(1,2,3)^T$ and $(1,1,3)^T$?

3. Are the following families of vectors of $\mathbb{R}^4$ linearly independent:

   (a) $(1,-1,1,-1)^T, (1,1,2,-2)^T, (1,5,4,-4)^T$,

   (b) $(1,-1,1,-1)^T, (1,1,2,-2)^T, (4,2,7,-2)^T$.

4. Determine whether the following sets of vectors are bases for $\mathbb{R}^3$:

   (a) $(1,0,1)^T, (1,1,0)^T, (0,1,1)^T, (2,1,1)^T$,

   (b) $(1,0,1)^T, (1,1,0)^T, (1,2,-1)^T$,

   (c) $(1,0,1)^T, (0,1,1)^T, (1,2,-1)^T$.

Problems.

5. **New linearly independent families from old ones.** Let $\vec{u}, \vec{v}, \vec{w}$ be linearly independent vectors of $\mathbb{R}^n$.

   (a) Show that the vectors $\vec{u}+\vec{v},\ \vec{v}+\vec{w},\ \vec{w}+\vec{u}$ are also linearly independent.

   (b) Are the vectors $\vec{u}-\vec{v},\ \vec{v}-\vec{w},\ \vec{w}-\vec{u}$ linearly independent?

6. **An explicit description of a span.** Consider the plane of $\mathbb{R}^3$ of equation

$$H := \{(x,y,z)^T \mid x+y=z\}$$

   (a) Show that $\vec{u} := (1,0,1)^T$ and $\vec{v} := (0,1,1)^T$ belong to H.

   (b) Show that $\mathrm{span}(\vec{u},\vec{v})$ is contained in H.

   (c) Show that $\mathrm{span}(\vec{u},\vec{v}) = H$.

7. **The incomplete basis theorem.** Let $\vec{v}_1, \ldots, \vec{v}_k$ be a linearly independent family of vectors of $\mathbb{R}^n$.

   (a) Let $\vec{v} \in \mathbb{R}^n$ be a vector that is not contained in $\text{span}(\vec{v}_1, \ldots, \vec{v}_k)$. Show that the family $\vec{v}_1, \ldots, \vec{v}_k, \vec{v}$ is still linearly independent.

   (b) By applying the previous question several times if necessary, show that it is possible to complete the family $\vec{v}_1, \ldots, \vec{v}_k$ into a basis $\vec{v}_1, \ldots, \vec{v}_n$ of $\mathbb{R}^n$.

8. **Extracting a basis.**

   (a) Let $\vec{v}_1, \ldots, \vec{v}_k$ be a family of vectors of $\mathbb{R}^n$ with $k \geq 2$. Show that if $\vec{v}_k \in \text{span}(\vec{v}_1, \ldots, \vec{v}_{k-1})$, then $\text{span}(\vec{v}_1, \ldots, \vec{v}_k) = \text{span}(\vec{v}_1, \ldots, \vec{v}_{k-1})$.

   (b) Let $\vec{v}_1, \ldots, \vec{v}_k$ be a spanning family of vectors of $\mathbb{R}^n$. Show that we can extract a sub-family that forms a basis of $\mathbb{R}^n$.